

Predicting Chaos Statistically: The Double Pendulum

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The motivating question behind this project was whether predictions can be made about chaotic systems. In particular, it was expected that certain statistical methods may be useful in order to make long term predictions on average for certain phenomena. To study this, the project attempted determine whether or not, for a double pendulum, the frequency of flips, the state in which the lower pendulum has an angle of 180 degrees with respect to the vertical, depends on initial conditions or only on the energy of the system. This was tested by a time evolution of the equations of motion, which was then compared to a calculation based on an integral over the phase space of a family of solutions. For the tested energy, there was a 10 percent variation between initial conditions, and the discrepancy between the time evolutions and the calculated values ranged from 3 to 9 percent for different energy levels.

I. INTRODUCTION

The field of Analytical Mechanics teaches us how to study exact solutions to mechanical systems. However, most mechanical systems do not admit exact solutions arbitrarily far into the future, namely they are chaotic. In other words, two identical systems starting at very close initial conditions will become completely uncorrelated in their motion after some typical time (time to chaos).

One may wonder whether any long term predictions could be made for chaotic system. Since the motion is chaotic and it samples a considerable part of the allowed phase space, often all of it, one may expect statistical methods to work, namely to be able to make long time predictions on average.

In this work we chose one of the simplest chaotic systems, the double pendulum, in order to test this idea. The double pendulum is a system consisting of two pendulums, the first attached to the some juncture such as the ceiling, and the other positioned so that the top of the lower pendulum is attached to the bottom of the upper pendulum.

For concreteness, we chose to study the rate of flips as the object of prediction, where a flip event is defined to be when $\theta_2 = \pi \pmod{2\pi}$. The primary source of interest in these flips is that the frequency of the flips should only depend on the energy of the system, and not on the initial conditions, despite the chaotic nature of the double pendulum system.

Two methods were used to try to test this theory. The first involved using the Mathematica software to precisely solve the equations of motion for exact initial conditions, and then counting the number of flips over a period of time. The second involved solving an integral over the phase space, to find the frequency with which systems of a particular energy are at the state of a flip. The former method will be referred to as the simulation method, and the latter as the ensemble method.

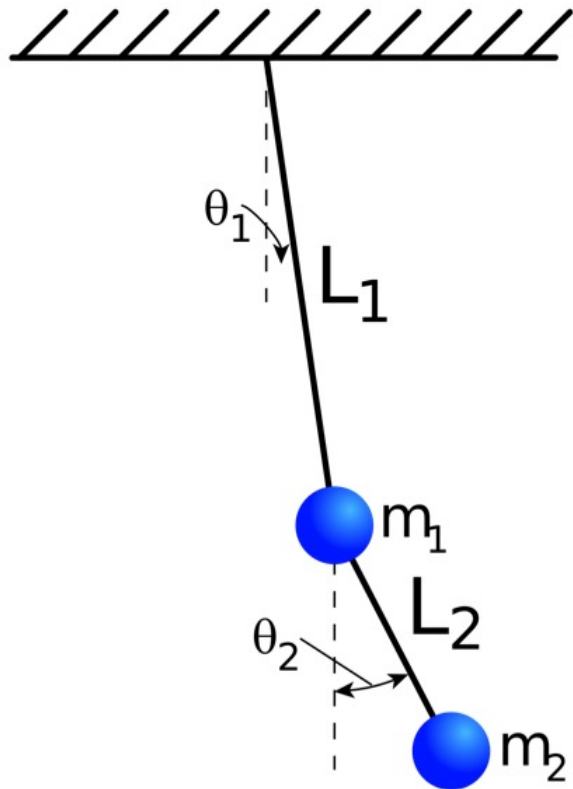


FIG. 1. Diagram of a double pendulum

II. SIMULATION METHOD

The simulation method method for finding the frequency of flips allowed for testing of many different initial conditions, because it relies on solving the equations of motions for a given set of initial conditions. Defining the generalized variables, θ_1 and θ_2 , as the angle between the upper pendulum and the vertical and between the lower pendulum and the vertical respectively, taking the

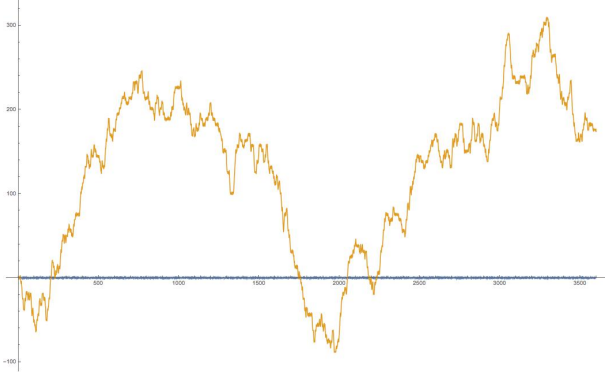


FIG. 2. Data for initial conditions of $\theta_1=\theta_2=\pi/2$, $\theta'_1=\theta'_2=0$. Plot of the angles θ_1 (blue) and θ_2 (orange) in radians vs. time in seconds

lengths of both pendulums to be 1 m, or to be more general one unit of length for any scaling, and the masses to be equal, in this case taken to be 1 kg, then the kinetic energy is given by

$$T = \theta_1'^2 + 0.5\theta_2'^2 + \theta_1'\theta_2'Cos(\theta_1 - \theta_2) \quad (1)$$

Setting the potential energy to be 0 at the horizontal line through the top of the upper pendulum, such that when $\theta_1 = \theta_2 = \pi/2$, $U=3g$, then

$$U = -g(2Cos(\theta_1) + Cos(\theta_2)) + 3 \quad (2)$$

Therefore the Lagrangian is

$$L = \theta_1'^2 + 0.5\theta_2'^2 + \theta_1'\theta_2'Cos(\theta_1 - \theta_2) + g(2Cos(\theta_1) + Cos(\theta_2) + 3) \quad (3)$$

The Euler-Lagrange equations are

$$\begin{aligned} \theta_1'' &= -(g/2)Sin(\theta_1) - (1/2)\theta_2''Cos(\theta_1 - \theta_2) - (1/2)\theta_2'^2Sin(\theta_1 - \theta_2) \\ \theta_2'' &= -(g/2)Sin(\theta_2) - \theta_1''Cos(\theta_1 - \theta_2) + \theta_1'^2Sin(\theta_1 - \theta_2) \end{aligned} \quad (4)$$

These equations were solved, and evolved over 3600 seconds using Mathematica with the NDSolve command. This was done for five different initial conditions for $E=0$.

A flip occurs at each point where $\theta_2=(2n+1)\pi$, where n is any integer, both positive and negative. Mathematics was used to locate all of these points, in order to determine when each flip occurred. The total number of flips was then counted in order to find the frequency. Because we expect that the frequency should not depend on initial conditions, the rate of flips should converge to the frequency over a long period of time. This was examined by plotting the average rate of flips per minute after each minute for each trial.

Taking the average rate from the different initial conditions yields a frequency of .193 flips/second for $E=3g$.

Individual simulations were conducted for $E=3.5g$ and $E=3.75g$ as well, with the following results.

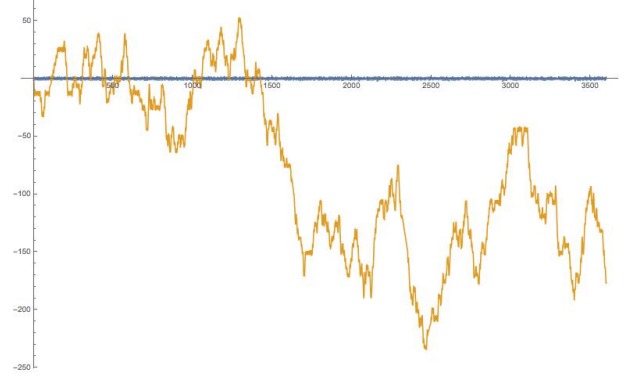


FIG. 3. Data for initial conditions of $\theta_1=-\pi/2, \theta_2=\pi/2$, $\theta'_1=\theta'_2=0$. Plot of the angles θ_1 (blue) and θ_2 (orange) in radians vs. time in seconds

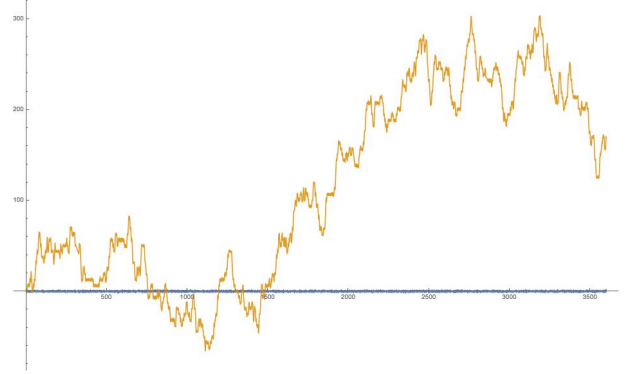


FIG. 4. Data for initial conditions of $\theta_1=2\pi/3, \theta_2=0$, $\theta'_1=\theta'_2=0$. Plot of the angles θ_1 (blue) and θ_2 (orange) in radians vs. time in seconds

III. ENSEMBLE METHOD

The frequencies found in the simulation method should be consistent with the expected value of the frequency

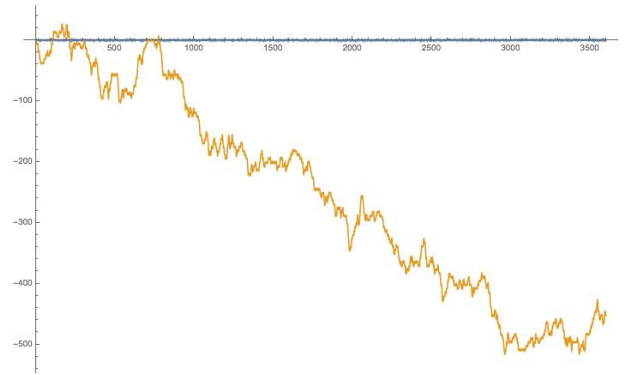


FIG. 5. Data for initial conditions of $\theta_1=\theta_2=\pi/3$, $\theta'_1=-1.565$, $\theta'_2=-1.935$ radians per second. Plot of the angles θ_1 (blue) and θ_2 (orange) in radians vs. time in seconds

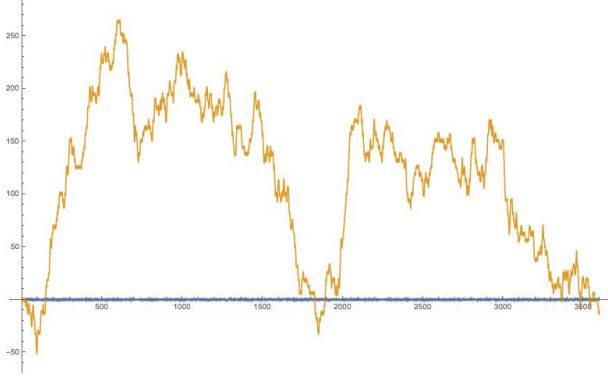


FIG. 6. Data for initial conditions of $\theta_1=\theta_2=0$, $\theta'_1=0$, $\theta'_2=5.422$ radians per second. Plot of the angles θ_1 (blue) and θ_2 (orange) in radians vs. time in seconds

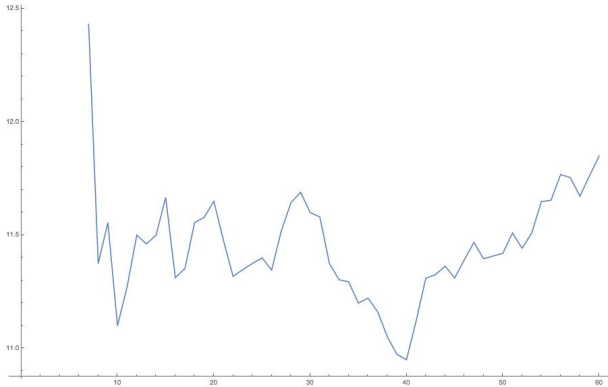


FIG. 7. Mean flip rate (flips per minute) vs. time (mins); $\theta_1=2\pi/3$ $\theta_2=0$, rate = 11.85 flips/min, 0.1975 flips/s

calculated as

$$\frac{\iiint |\theta'_2| \delta(H-E) \delta(\theta_2-\pi) dp_1 dp_2 d\theta_1 d\theta_2}{\iiint \delta(H-E) dp_1 dp_2 d\theta_1 d\theta_2} \quad (5)$$

The integral over the generalized momenta can be performed analytically to account for the Dirac delta func-

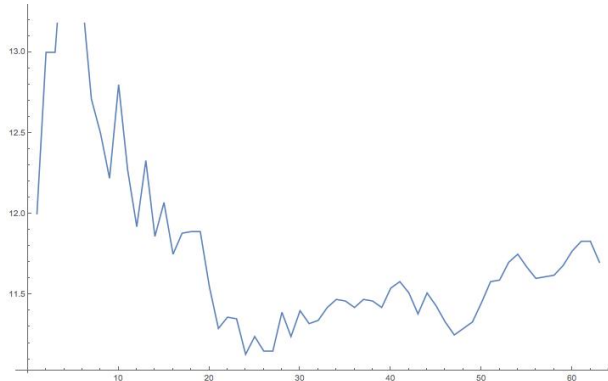


FIG. 8. Mean flip rate (flips per minute) vs. time (mins); $\theta_1=\theta_2=\pi/2$, rate = 11.7 flips/min, 0.195 flips/s

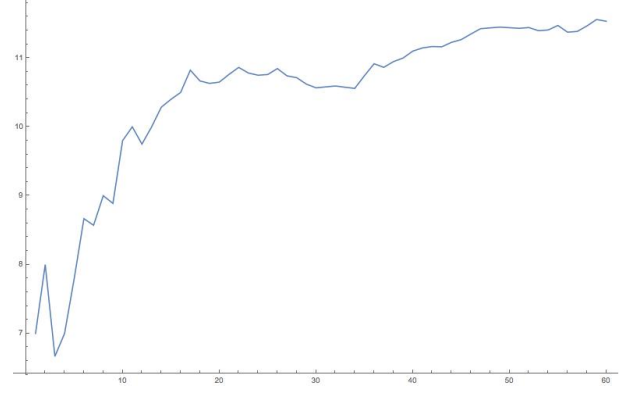


FIG. 9. Mean flip rate (flips per minute) vs. time (mins); $\theta_1=-\pi/2$ $\theta_2=\pi/2$, rate = 11.5333 flips/min, 0.19222 flips/s

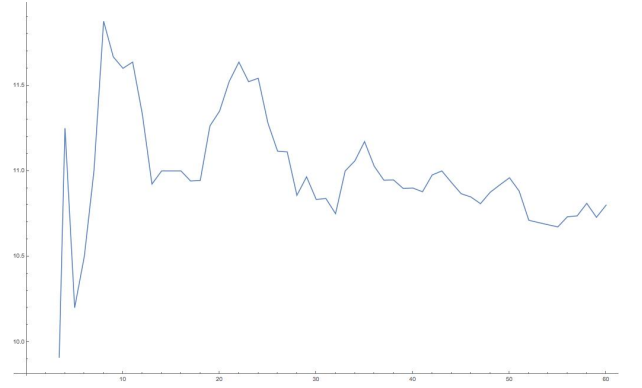


FIG. 10. Mean flip rate (flips per minute) vs. time (mins); $\theta_1=\pi/3$ $\theta_2=\pi/3$ $\theta'_1=-1.565$ $\theta'_2=-1.935$; rate = 10.8 flips/min, 0.18 flips/s

Energy	Rate(flips/second)
3.0g	.193
3.5g	.234
3.75g	.256

TABLE I. Flip frequency for each energy

tion. For $E=3g$, this results in an expression for the numerator of

$$\int ((2/3) - 4\cos^2(\theta) + (8/3)\cos^3(\theta) - (2/3) - (16/3) (\csc(\theta)\sin^2(\theta/2)\sin(2\theta))^{1.5}) / (\cos(\theta)-2) d\theta \quad (6)$$

with θ ranging from $-\pi/3$ to $\pi/3$.

For the denominator, the result still requires integrating over both generalized coordinates, and is

$$\iint 6\sqrt{(2\cos(\theta_1) - 4\cos^2(\theta_1) - 4\cos(\theta_1)\cos(\theta_2) -$$

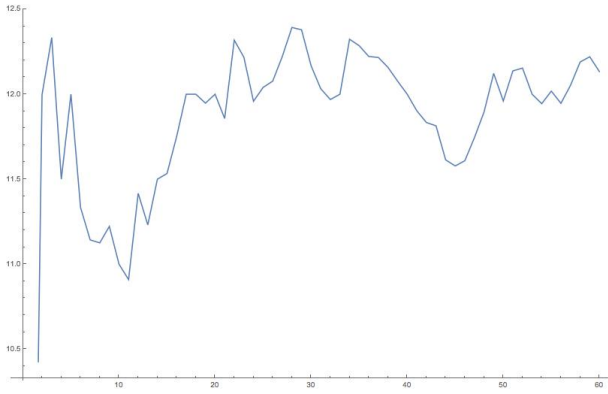


FIG. 11. Mean flip rate (flips per minute) vs. time (mins); $\theta_1=\theta_2=0$ $\theta'_1=0$ $\theta'_2=5.422$; rate = 12.1333 flips/min, 0.2022222 flips/s

$$\cos^2(\theta_2) + \cos(\theta_2)) d\theta_1 d\theta_2(7)$$

θ_1 ranges from $-2\pi/3$ to $2\pi/3$, and θ_2 ranges from $-\pi$ to π .

Solving these integrals using Mathematica produces $5.784/30.877 = .187$ flips per second. This varies by 3 percent from the frequency calculated via the simulation method.

The same integral was calculated for $E=0.5g$ and $E=0.75g$, and compared to the simulation values.

Energy	Simulation Rate	Ensemble Rate	Percent Difference
0	.193	.187	3
0.5g	.234	.213	9
0.75g	.256	.263	3

TABLE II. Comparison between the frequency calculated via the simulation and ensemble methods

IV. DISCUSSION

The results of these tests suggest that the frequency of flips does depend on the energy of the system rather than on initial conditions. The range of values for the frequency for simulations at $E=3.0g$ were within a 10 percent range, as were the values calculated via the ensemble method for each of the energies. However, it is unclear whether the discrepancy was caused by error, or due to real variation based on initial conditions. One likely cause for the discrepancy is the inclusion of motion shortly after the initial conditions, when the motion is not completely random. This may have impacted the frequency of flips early in the simulation, leading to small variation when averaged over a longer period. This may have been avoided by instead looking at a period of time once the motion has become random. Additionally, only one simulation was conducted for $E=3.5g$ and $E=3.75g$, so it is uncertain whether or not all energies have consistent frequencies of flips. Examining that question more fully could also lead to efforts to find a relation between the energy level and the frequency of flips.

Another explanation for the discrepancy is the assumption used in the ensemble method that the system is fully chaotic. In that case, the motion will fill the entire phase space, and therefore viewing a family of systems that cover the entire phase space is identical. However, for many energy levels, the double pendulum does not fill the entire phase space, instead there are islands of stable solutions that the random motion may not reach. In fact, the size of the discrepancy may give a clue as to how much of the phase space the motion fills for a given energy. The relative small size of the discrepancy suggests that the motion may fill up to 97 percent of the phase space, giving credibility to the use of the ensemble method.